

Perilesional MRE forward-and-inverse problem

3D vector Navier with tensor stiffness on the Sobh–Ehman phantoms

A step-by-step specification for the computational team

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Purpose.

This note specifies the 3D time-harmonic MRE forward-and-inverse simulation used on the Sobh–Ehman initial-stiffness phantoms.

The physics is set by what MRE actually measures — a **three-component displacement vector** $\mathbf{u}(\mathbf{x})$ inside a heterogeneous **anisotropic** elastic solid — so the forward model is the vector Navier equation with a rank-2 tensor stiffness field $\mu_{ij}(\mathbf{x})$ constructed by rotating the Sobh–Ehman principal-frame acoustoelastic moduli into Cartesian coordinates.

The inversion side follows Yin (2026): the curl of the vector displacement is taken, decomposed into 20 Fibonacci-sphere directions in k -space, direct-inverted per direction, and combined into a μ_{TSM} (voxelwise MIP) and μ_{conv} (amplitude-weighted mean).

Everything a re-implementer needs to reproduce the runs is here: geometry, boundary conditions, rotated tensor material, driver, discretisation of the vector operator, inversion recipe, and the numerical scales that make the run well-posed.

1 Physical picture

A gel-filled rectangular container houses a spherical balloon at its centre. The balloon has been inflated with water to 250 mL (radius $a = 3.908$ cm) from a stress-free casting state at 50 mL (radius $A = 2.285$ cm). A piston plate at the bottom of the container ($z = z_{\min}$) is driven upward at 80 Hz; the resulting shear-wave field is sampled inside the gel and inverted for the local shear modulus. This mimics the MRE experiment reported by Yin et al. (2026) as reproduced quantitatively by *paper_lesion_expansion.pdf*.

2 Governing equation: vector Navier with tensor stiffness

We solve the time-harmonic vector Navier equation for the complex displacement $\mathbf{u}(\mathbf{x}) \in \mathbb{C}^3$ at Yin’s phantom frequency $\omega = 2\pi f$, $f = 80$ Hz:

$$\boxed{\rho \omega^2 u_i(\mathbf{x}) = \partial_j \sigma_{ij}(\mathbf{x}) \quad \text{in } \Omega} \tag{1}$$

with a quasi-anisotropic constitutive law appropriate for the rotated tangential/radial acoustoelastic tensor stiffness $\mu_{ij}(\mathbf{x})$ (spelled out in §5):

$$\sigma_{ij} = \lambda \delta_{ij} \varepsilon_{kk} + \mu_{ik}(\mathbf{x}) \varepsilon_{kj} + \mu_{jk}(\mathbf{x}) \varepsilon_{ki}, \quad \varepsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i) \tag{2}$$

with hysteretic damping folded into the stiffness:

$$\mu_{ij}^*(\mathbf{x}) = \mu_{ij}(\mathbf{x}) (1 + i \xi), \quad \xi = 0.05 \quad (\Rightarrow Q \approx 10). \quad (3)$$

Physical constants:

Symbol	Value	Meaning
ρ	1000 kg m^{-3}	gel density
ω	$2\pi \cdot 80 \text{ rad s}^{-1}$	angular frequency (Yin's)
λ	$1 \times 10^7 \text{ Pa}$	bulk-penalty first Lamé (near-incompressible)
ξ	0.05	hysteretic damping ($Q \approx 10$)

3 Domain, grid and physical scales

The container is Yin's $15 \times 15 \times 18 \text{ cm}$ box: horizontal footprint 15 cm in each in-plane direction, vertical height 18 cm . The FD grid matches these dimensions exactly at 3 mm voxel resolution.

Symbol	Value	Meaning
(N_x, N_y, N_z)	$(50, 50, 60)$	voxels along x, y, z
Δx	3 mm	voxel edge (isotropic)
(L_x, L_y, L_z)	$(15, 15, 18) \text{ cm}$	container extents
DOF	$3 N_x N_y N_z = 450\,000$	unknowns in the linear system

Axis convention (used throughout this document and the code). Physical Cartesian axes (x, y, z) map to grid indices (i, j, k) identically: i indexes x , j indexes y , k indexes z . z is **vertical, increasing upward**: $k = 0$ is the bottom (where the driver piston pushes upward), $k = N_z - 1$ is the top (traction-free free-surface). The container centre (and balloon centre) sit at $((N_x - 1)/2, (N_y - 1)/2, (N_z - 1)/2)$. Grid arrays therefore have shape (N_x, N_y, N_z) or $(N_x, N_y, N_z, \dots)$ for tensor fields.

Shear-wave resolution. At 80 Hz in unstretched gel (using the P1 baseline $\mu_0 = 3530 \text{ Pa}$):

$$c_s = \sqrt{\mu_0 / \rho} = \sqrt{3530 / 1000} \approx 1.88 \text{ m s}^{-1} \quad \Rightarrow \quad \lambda_s = c_s / f_{\text{drive}} \approx 23.5 \text{ mm} \quad \Rightarrow \quad \lambda_s / \Delta x \approx 6.3. \quad (4)$$

That is more than the 6-per-wavelength rule of thumb; the FD stencil resolves the shear wave without aliasing.

4 Boundary conditions

The container has D_{4h} -broken BCs (five rigid walls, one traction-free top):

Region	Condition	Discrete implementation
$k = N_z - 1$ (top face, whole)	$\sigma_{ij}n_j = 0, \mathbf{n} = +\hat{z}$	ghost-node stencil (eq. 5)
$k = 0$ (bottom face, wall region)	$\mathbf{u} = \mathbf{0}$	Dirichlet
$j = 0, j = N_y - 1$ (wall region)	$\mathbf{u} = \mathbf{0}$	Dirichlet
$i = 0, i = N_x - 1$ (wall region)	$\mathbf{u} = \mathbf{0}$	Dirichlet
Central driver disk (each of 5 faces)	face-normal u_c prescribed	overrides the wall Dirichlet
Balloon interior	$\mu_{ij}(\mathbf{x}) = 1 \text{ Pa } \delta_{ij}$	water <i>fluidinclusion</i> (§4)

Each of the five non-top faces is rigid ($\mathbf{u} = \mathbf{0}$) except on a centred circular disk (§6) whose face-normal component is Dirichlet-prescribed to the driver amplitude. The disk condition takes precedence over the wall condition where they overlap.

Traction-free top BC (ghost-node stencil). The physically correct top-face BC is $\sigma_{ij}n_j = 0$ with $\mathbf{n} = +\hat{z}$ (the outward normal at $k = N_z - 1$), i.e. $\sigma_{c2}(i, j, N_z - 1) = 0$ for $c \in \{0, 1, 2\}$ (where index $c = 2$ denotes the z -component of the stress-divergence). This clamps the *normal stress*, not the normal strain (an important distinction: an earlier $\partial u / \partial z = 0$ ghost-mirror gave TSM/conv ≈ 1.32 by clamping strain, well below Yin’s measured 1.57; the correct $\sigma \cdot \mathbf{n} = 0$ recovers the ratio).

Under the locally-isotropic assumption at the top slab ($\mu_{ij} \approx \bar{\mu}_{\text{top}} \delta_{ij}$ — valid because the top face sits far from the pre-stressed balloon), the three conditions solve algebraically for the ghost values $u_c(i, j, N_z)$ (one slab above the top) in terms of the interior neighbours at $(i, j, N_z - 2)$, $(i \pm 1, j, N_z - 1)$, and $(i, j \pm 1, N_z - 1)$:

$$\begin{aligned}
 u_z(i, j, N_z) &= u_z(i, j, N_z - 2) - f_{\text{top}} \left[u_x(i+1, j, N_z - 1) - u_x(i-1, j, N_z - 1) \right. \\
 &\quad \left. + u_y(i, j+1, N_z - 1) - u_y(i, j-1, N_z - 1) \right] \\
 u_x(i, j, N_z) &= u_x(i, j, N_z - 2) - [u_z(i+1, j, N_z - 1) - u_z(i-1, j, N_z - 1)] \\
 u_y(i, j, N_z) &= u_y(i, j, N_z - 2) - [u_z(i, j+1, N_z - 1) - u_z(i, j-1, N_z - 1)]
 \end{aligned} \tag{5}$$

with

$$f_{\text{top}} = \frac{\lambda}{\lambda + 2\bar{\mu}_{\text{top}}}, \quad \bar{\mu}_{\text{top}} = \left\langle \frac{1}{3} \text{tr } \mu(\mathbf{x}) \right\rangle_{k=N_z-1} = \frac{1}{N_x N_y} \sum_{i,j} \frac{1}{3} \text{tr } \mu(i, j, N_z - 1). \tag{6}$$

Note $\bar{\mu}_{\text{top}}$ is a *single scalar*: the spatial mean of $\frac{1}{3} \text{tr } \mu$ over the entire top face slab $k = N_z - 1$ (`vector_elasticity_3d.py`). This makes f_{top} a constant across the top face, so the ghost coefficients in (5) are (i, j) -independent. Legitimate because the top face sits far outside the pre-stressed region where $\mu_{ij}(\mathbf{x})$ has already relaxed to the near-isotropic background value.

The minus signs in (5) are physical: because the ghost sits *above* the interior (larger z), the central-FD sign $[u(N_z) - u(N_z - 2)] / (2\Delta x)$ produces the opposite sign convention from the case where the ghost sat below.

No ghost DOF is introduced: every interior-stencil access to $u_c(i, j, N_z)$ is distributed across the RHS of (5) in the sparse assembly.

Balloon: water — incompressible fluid inclusion. The balloon is treated as a single water inclusion with:

- **No shear support:** $\mu_{\text{ball}} \approx 0$ (numerically 1 Pa, essentially zero — water is a fluid and cannot support shear stress at any timescale).
- **High bulk stiffness:** enforced by the global bulk-penalty first Lamé $\lambda = 1 \times 10^7$ Pa = 10 MPa. Real water $K \approx 2.2$ GPa — our 10 MPa is 220× softer, but at 80 Hz the P-wavelength inside the ball is $\sqrt{\lambda/\rho}/f \approx 1.25$ m, seven times the container extent, so P-waves inside the ball are quasi-static — pressure equilibrates instantly relative to the shear timescale. Increasing λ further gives no wave-field change in the perilesional shell.

Note that $\lambda = 1 \times 10^7$ Pa is a *global* bulk penalty — it applies everywhere, gel and ball alike. This is physically justified: real gel is also quasi-incompressible ($K \sim$ GPa), close to water. Both media are near-incompressible; they differ only in shear behaviour, which is what MRE actually measures.

The balloon is *attached* to the surrounding gel: the gel was cast against the balloon surface, so at $r = a$ the FD stencil carries displacement continuity $\mathbf{u}_{\text{gel}}(a^+) = \mathbf{u}_{\text{ball}}(a^-)$ automatically (single voxel field, material coefficients simply jump across the interface).

What this model implies for the wave field. Inside the ball, the wave equation reduces to $\rho\omega^2 u_i \approx \partial_i(\lambda \partial_k u_k)$ (P-only, no shear). Shear waves cannot propagate through the ball; they meet a fluid interface at $r = a$ where the tangential traction must be zero. The ball then behaves as a near-rigid *coupled* inclusion at the wave frequency — not because the water is stiff (it isn't in shear), but because water's incompressibility plus the attached-interface constraint prevents the ball from deforming tangentially to accommodate the varying gel wave field. Small tangential slip at the surface accommodates the geometric over-constraint that arises when the gel wavelength (~ 25 mm) is smaller than the ball diameter (~ 78 mm).

5 Material field $\mu_{ij}(\mathbf{x})$

5.1 Kinematics: incompressible spherical cavity expansion

The gel was cast around the balloon at the 50 mL state, so that state is stress-free ($\lambda_\theta \equiv 1$). Incompressibility of the gel combined with spherical symmetry gives the exact isochoric radial map from reference radius R to current radius r :

$$r^3 = R^3 + (a^3 - A^3), \quad \lambda_\theta = \frac{r}{R} = \left(1 + \frac{a^3 - A^3}{R^3}\right)^{1/3}, \quad \lambda_r = \lambda_\theta^{-2}, \quad (7)$$

with the current cavity radius set by injection volume, $a = (3V/(4\pi))^{1/3}$. **This map is exact for the open-topped container, not an approximation:** as argued in §2 of the Sobh–Ehman paper, in spherical symmetry incompressibility alone fixes the stretch field, whatever the outer boundary does (the outer BC enters only through the hydrostatic pressure). For $V = 250$ mL, $a = 3.908$ cm; λ_θ at the cavity edge is 1.710 and at $r = a + 12$ mm is 1.159 (matches paper Table 1).

5.2 Acoustoelastic (perilesional) modulus

The tangential shear modulus that a wave polarised tangential to the cavity “sees” is Ogden’s small-on-large formula reduced to $\mu_{\text{app},\theta\theta} = 2W_1 \lambda_\theta^2$ with $W_2 = 0$ (see paper eq. 4). We use the **fitted Sobh–Ehman material model** (`perilesional_model/sce_models.py`), which pins the baseline μ_0 to the digitised Fig. 6 series at $\lambda_\theta = 1$ and identifies a P2 fibre correction from the P1-normalised ratio:

Constitutive laws — fitted material.

Both phantoms share the same functional form:

$$\mu_{\text{app},\theta\theta}(\mathbf{x}) = \mu_0 f(I_1) \lambda_\theta^2, \quad I_1(\mathbf{x}) = \lambda_\theta^{-4} + 2 \lambda_\theta^2, \quad (8)$$

with I_1 the first invariant under incompressible spherical kinematics ($\lambda_r = \lambda_\theta^{-2}$).

P1 (plain 10 % bovine gelatin, neo-Hookean matrix):

$$f^{\text{P1}}(I_1) \equiv 1 \quad \Rightarrow \quad \mu_{\text{app},\theta\theta}^{\text{P1}} = \mu_0^{\text{P1}} \lambda_\theta^2. \quad (9)$$

P2 (8 % gelatin + 7 % cellulose fibre, matrix + power-law fibre recruitment):

$$f^{\text{P2}}(I_1) = 1 + c(I_1 - 3)^m, \quad c = 0.2728, \quad m = 0.5791. \quad (10)$$

Fitted material constants (`sce_models.py`):

Symbol	Value	Units	Meaning
μ_0^{P1}	3530	Pa	P1 baseline (Fig. 6 pinned at $\lambda_\theta = 1$)
μ_0^{P2}	3340	Pa	P2 baseline (same pinning)
c	0.2728	—	P2 power-law fibre recruitment coefficient
m	0.5791	—	P2 power-law fibre recruitment exponent

The scalar stiffening factor $f(I_1)$ enters both principal wave moduli identically (Ogden’s small-on-large with $W_2 = 0$): $\mu_{\theta\theta} = \mu_0 f(I_1) \lambda_\theta^2$, $\mu_{rr} = \mu_0 f(I_1) \lambda_\theta^{-4}$. For P1, $f \equiv 1$ recovers the plain neo-Hookean case; the P2 power-law form was selected because it fits the shell-averaged data across all five inflation volumes with just two parameters, and its kinematic factor $g = I_1 - 3$ ensures $f \rightarrow 1$ at the reference state.

5.3 Rotation to Cartesian — the tensor field $\mu_{ij}(\mathbf{x})$

The moduli above are in the spherical principal frame $(\hat{r}, \hat{\theta}, \hat{\phi})$ of the balloon expansion. The FD grid is in Cartesian coordinates (x, y, z) , so we rotate the tensor at every voxel. This subsection derives the change-of-basis formula in full and shows how the tangential-eigenvalue degeneracy of this problem collapses it to a form that requires only $\hat{r}$.

General transformation. In the spherical principal frame the tensor is diagonal with the Ogden eigenvalues on its diagonal:

$$\mu_{\text{sph}} = \begin{pmatrix} \mu_{rr} & 0 & 0 \\ 0 & \mu_{\theta\theta} & 0 \\ 0 & 0 & \mu_{\phi\phi} \end{pmatrix}, \quad \mu_{rr} = 2W_1 \lambda_\theta^{-4}, \quad \mu_{\theta\theta} = \mu_{\phi\phi} = 2W_1 \lambda_\theta^2. \quad (11)$$

Let $Q(\mathbf{x})$ be the rotation matrix whose columns are the local spherical basis vectors expressed in Cartesian components:

$$Q(\mathbf{x}) = [\hat{r}(\mathbf{x}) \mid \hat{\theta}(\mathbf{x}) \mid \hat{\phi}(\mathbf{x})] \in \text{SO}(3), \quad QQ^\top = I, \quad \det Q = +1. \quad (12)$$

With colatitude θ measured from $+\hat{z}$ and azimuth ϕ from $+\hat{x}$:

$$\hat{r} = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}, \quad \hat{\theta} = \begin{pmatrix} \cos \theta \cos \phi \\ \cos \theta \sin \phi \\ -\sin \theta \end{pmatrix}, \quad \hat{\phi} = \begin{pmatrix} -\sin \phi \\ \cos \phi \\ 0 \end{pmatrix}. \quad (13)$$

A rank-2 tensor transforms under a rigid rotation as $\mu' = Q \mu Q^\top$, so the Cartesian tensor field is

$$\boxed{\mu_{\text{Cart}}(\mathbf{x}) = Q(\mathbf{x}) \mu_{\text{sph}}(\mathbf{x}) Q^\top(\mathbf{x})} \quad (14)$$

or in index form $\mu_{ij}(\mathbf{x}) = Q_{i\alpha}(\mathbf{x}) \mu_{\text{sph},\alpha\beta}(\mathbf{x}) Q_{j\beta}(\mathbf{x})$ (Greek indices $\alpha, \beta \in \{r, \theta, \phi\}$; Latin $i, j \in \{x, y, z\}$).

Degeneracy reduction. Expanding (14) column-by-column with $\mu_{\text{sph}} = \text{diag}(\mu_{rr}, \mu_{\theta\theta}, \mu_{\phi\phi})$:

$$\mu_{\text{Cart}} = \mu_{rr} \hat{r} \hat{r}^\top + \mu_{\theta\theta} \hat{\theta} \hat{\theta}^\top + \mu_{\phi\phi} \hat{\phi} \hat{\phi}^\top. \quad (15)$$

In this problem $\mu_{\theta\theta} = \mu_{\phi\phi}$ (there is no azimuthal preference — both tangential eigenvalues come from the same $2W_1\lambda_\theta^2$). The last two terms of (15) factor:

$$\mu_{\text{Cart}} = \mu_{rr} \hat{r} \hat{r}^\top + \mu_{\theta\theta} (\hat{\theta} \hat{\theta}^\top + \hat{\phi} \hat{\phi}^\top). \quad (16)$$

The orthonormality of Q (12) gives the completeness relation $\hat{r} \hat{r}^\top + \hat{\theta} \hat{\theta}^\top + \hat{\phi} \hat{\phi}^\top = I$, so $\hat{\theta} \hat{\theta}^\top + \hat{\phi} \hat{\phi}^\top = I - \hat{r} \hat{r}^\top$. Substituting:

$$\boxed{\mu_{ij}(\mathbf{x}) = \mu_{rr}(r) \hat{r}_i(\mathbf{x}) \hat{r}_j(\mathbf{x}) + \mu_{\theta\theta}(r) (\delta_{ij} - \hat{r}_i(\mathbf{x}) \hat{r}_j(\mathbf{x}))} \quad (17)$$

with $\hat{r}(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_{\text{cavity}})/r$ the Cartesian components of the local radial unit vector. The tangent-plane basis $(\hat{\theta}, \hat{\phi})$ has dropped out entirely through the projector $(I - \hat{r} \hat{r}^\top)$; only $\hat{r}$ survives. This is why the assembly code (`paper_wave_sim_sobh_vector_mip_multiface.py`:99–102) never builds $\hat{\theta}, \hat{\phi}$ or the full Q .

Gauge remark. The choice of $(\hat{\theta}, \hat{\phi})$ inside the tangent plane is a *gauge freedom* of the eigenvalue-degenerate problem: any orthonormal frame in the tangent plane rotates μ_{sph} 's lower 2×2 block into itself, so (14) is invariant under those in-plane rotations. If the two tangential eigenvalues were split (e.g. by an explicit azimuthal fiber preference), that gauge would be physically fixed and one would need $\hat{\theta}, \hat{\phi}$ explicitly.

Equation (17) gives a **fully populated symmetric 3×3 tensor field**: six independent components varying voxel-by-voxel.

On-axis check. At the $+\hat{z}$ pole $\hat{r} = (0, 0, 1)^\top$, so $\hat{r}\hat{r}^\top = \text{diag}(0, 0, 1)$ and (17) reads

$$\mu_{\text{Cart}}(\text{pole}) = \text{diag}(\mu_{\theta\theta}, \mu_{\theta\theta}, \mu_{rr}). \quad (18)$$

Radial stiffness lands on the z -diagonal (perpendicular to the local balloon surface); tangential stiffness lands on the x, y diagonals (in the tangent plane). Consistent with intuition.

Axis convention. The derivation uses the standard Cartesian convention (x, y, z) with colatitude θ measured from $+\hat{z}$. The implementation matches this exactly: grid-index ordering $(i, j, k) \leftrightarrow (x, y, z)$ (see §3), so $\hat{r}$ is stacked as $(\hat{r}_x, \hat{r}_y, \hat{r}_z)$ in the code, and all Cartesian tensor components μ_{ij}, C_{ijkl} inherit that same (x, y, z) order.

Anisotropy magnitude. At the cavity edge at 250 mL ($\lambda_\theta = 1.71$, $I_1 = 5.97$, $g = 2.97$, $f^{\text{P2}} = 1.516$):

Component	Formula	P1 [kPa]	P2 [kPa]	Ratio to $\mu_{\theta\theta}$
$\mu_{\theta\theta} = \mu_{\phi\phi}$	$\mu_0 f(I_1) \lambda_\theta^2$	10.32	14.77	1.00
μ_{rr}	$\mu_0 f(I_1) \lambda_\theta^{-4}$	0.41	0.59	0.040

The ring is **25× stiffer in the tangential direction than in the radial direction** — irreducible anisotropy, not a small correction. A scalar $G(\mathbf{x})$ representation cannot capture it, which is one of the reasons the forward model must be (1). The P2 fibre boost ($\sim 1.5\times$ over P1) sits on top of the same anisotropic structure.

5.4 Solver inputs: complete summary

Everything the vector-Navier solver reads at each voxel is derived from a small set of scalar constants plus the incompressible spherical kinematics. Consolidated here for reproducibility.

Scalar inputs (uniform across the domain).

Symbol	Value	Units	Meaning
μ_0^{P1}	3530	Pa	P1 baseline shear modulus (Fig. 6 pinned)
μ_0^{P2}	3340	Pa	P2 baseline shear modulus (Fig. 6 pinned)
c	0.2728	—	P2 power-law fibre recruitment coefficient
m	0.5791	—	P2 power-law fibre recruitment exponent
λ	10^7	Pa	first Lamé — near-incompressibility penalty
ρ	1000	kg m^{-3}	gel density
ξ	0.05	—	hysteretic damping ($Q \approx 10$)
f_{drive}	80	Hz	Yin’s phantom frequency
μ_{water}	1	Pa	balloon-interior placeholder (no shear support)

Kinematic geometry (drives $\lambda_\theta(r)$).

Symbol	Value	Units	Meaning
A	22.85	mm	reference balloon radius (50 mL casting)
a	39.08	mm	current balloon radius (250 mL inflated)

Per-voxel material fields. Two spherical-frame eigenvalues are computed from $\lambda_\theta(r)$ and a per-phantom stiffening factor $f(I_1)$:

- **P1 (neo-Hookean matrix):** $f^{P1}(I_1) \equiv 1$, so $W_1^{P1} = \mu_0^{P1}/2 = 1765$ Pa (constant).
- **P2 (matrix + power-law fibre recruitment):** $W_1^{P2}(\lambda_\theta) = (\mu_0^{P2}/2) [1 + c(I_1 - 3)^m]$ with $I_1 = \lambda_\theta^{-4} + 2\lambda_\theta^2$ from incompressible spherical kinematics.

Then $\mu_{\theta\theta} = 2W_1\lambda_\theta^2$, $\mu_{rr} = 2W_1\lambda_\theta^{-4}$, rotated to Cartesian via (17), with hysteretic damping folded in as $\mu_{ij}^* = \mu_{ij}(1 + i\xi)$.

Concrete values at representative radii.

Location	λ_θ	$\mu_{\theta\theta}^{P1}$ (kPa)	μ_{rr}^{P1} (kPa)	$\mu_{\theta\theta}^{P2}$ (kPa)	μ_{rr}^{P2} (kPa)
Cavity edge $r = a$	1.71	10.32	0.41	14.77	0.59
Ring mid $r = a + 12$ mm	1.16	4.75	1.95	5.04	2.07
Far field $r \gg a$	1.00	3.53	3.53	3.34	3.34
Balloon interior	—	0.001	0.001	0.001	0.001

The $25\times$ radial-vs-tangential anisotropy at the cavity edge is what drives the direction-selective wave response Yin’s TSM MIP picks out. P2 is $\sim 1.5\times$ stiffer tangentially than P1 in the shell — the power-law fibre recruitment kicking in.

Full 6×6 Voigt stiffness matrix at representative voxels. The rank-2 shear-modulus tensor $\mu_{ij}(\mathbf{x})$ and the Lamé constant λ together determine the full rank-4 elasticity tensor via

$$C_{ijkl}(\mathbf{x}) = \lambda \delta_{ij} \delta_{kl} + \frac{1}{2} (\mu_{ik} \delta_{jl} + \mu_{il} \delta_{jk} + \mu_{jk} \delta_{il} + \mu_{jl} \delta_{ik}) \quad (19)$$

which in the isotropic limit $\mu_{ij} = \mu \delta_{ij}$ recovers the standard $C_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$ with the two Lamé constants. Equation (19) is the specific rank-4 tensor the solver applies at every voxel.

Numerical verification against Ogden. At the cavity edge in the principal frame, C_{ijkl} computed from (19) with $\mu_{ij} = \text{diag}(\mu_{rr}, \mu_{\theta\theta}, \mu_{\theta\theta})$ matches the Ogden small-on-large elasticity tensor A_{ijkl}^0 (from linearising the incompressible neo-Hookean $W = \frac{\mu_0}{2}(I_1 - 3)$ around the pre-deformed state) to machine precision. All six diagonal entries agree, the off-diagonal λ entries agree, and the three shear moduli $(\mu_{rr} + \mu_{\theta\theta})/2$ and $(\mu_{\theta\theta} + \mu_{\phi\phi})/2$ agree. So the constitutive law encoded in (19) is the canonical continuum-mechanical form for our material — not an ad-hoc simplification.

Voigt convention (with the code’s axis ordering $(0, 1, 2) = (x, y, z)$, standard physics convention):

$$I=1 \rightarrow xx, \quad I=2 \rightarrow yy, \quad I=3 \rightarrow zz, \quad I=4 \rightarrow yz, \quad I=5 \rightarrow xz, \quad I=6 \rightarrow xy.$$

The 6×6 matrix at four representative voxels for each phantom is shown below — showing both the phantom's near-cavity anisotropy and the far-field near-isotropy. All values in kPa; $\lambda = 100$ kPa everywhere. Voigt indices follow the (x, y, z) ordering above. Each matrix is followed by its six eigenvalues, which are basis-invariant.

Phantom P1 (neo-Hookean, $\mu_0 = 3530$ Pa)

Cavity edge, $-z$ axis $(i, j, k) = (24, 24, 15)$, $r = 43.6$ mm

	xx	yy	zz	yz	xz	xy
xx	112.54	100.00	100.00	0.00	-0.18	-0.01
yy	100.00	112.54	100.00	-0.18	0.00	-0.01
zz	100.00	100.00	102.26	-0.18	-0.18	0.00
yz	0.00	-0.18	-0.18	3.70	0.00	-0.09
xz	-0.18	0.00	-0.18	0.00	3.70	-0.09
xy	-0.01	-0.01	0.00	-0.09	-0.09	6.27

eigenvalues (kPa): {3.68, 3.70, 5.61, 6.28, 12.54, 309.19}

Mid-shell off-axis $(i, j, k) = (31, 29, 17)$, $r = 44.4$ mm

	xx	yy	zz	yz	xz	xy
xx	110.13	100.00	100.00	0.00	1.77	-0.64
yy	100.00	111.08	100.00	1.22	0.00	-0.64
zz	100.00	100.00	105.17	1.22	1.77	0.00
yz	0.00	1.22	1.22	4.06	-0.32	0.88
xz	1.77	0.00	1.77	-0.32	3.83	0.61
xy	-0.64	-0.64	0.00	0.88	0.61	5.30

eigenvalues (kPa): {2.99, 3.90, 5.44, 7.44, 10.98, 308.84}

Far corner (+, +, +) $(i, j, k) = (47, 47, 57)$, $r = 126.2$ mm

	xx	yy	zz	yz	xz	xy
xx	107.08	100.00	100.00	0.00	-0.06	-0.05
yy	100.00	107.08	100.00	-0.06	0.00	-0.05
zz	100.00	100.00	107.03	-0.06	-0.06	0.00
yz	0.00	-0.06	-0.06	3.53	-0.02	-0.03
xz	-0.06	0.00	-0.06	-0.02	3.53	-0.03
xy	-0.05	-0.05	0.00	-0.03	-0.03	3.54

eigenvalues (kPa): {3.48, 3.55, 3.57, 7.05, 7.08, 307.06}

Balloon interior (water fluidinclusion, $\mu_{\text{ball}} = 10^6$ Pa) $(i, j, k) = (24, 24, 29)$, $r = 2.6$ mm

	xx	yy	zz	yz	xz	xy
xx	2100.00	100.00	100.00	0.00	0.00	0.00
yy	100.00	2100.00	100.00	0.00	0.00	0.00
zz	100.00	100.00	2100.00	0.00	0.00	0.00
yz	0.00	0.00	0.00	1000.00	0.00	0.00
xz	0.00	0.00	0.00	0.00	1000.00	0.00
xy	0.00	0.00	0.00	0.00	0.00	1000.00

eigenvalues (kPa): {1000.00, 1000.00, 1000.00, 2000.00, 2000.00, 2300.00}

Phantom P2 (matrix + power-law fibre, $\mu_0 = 3340$ Pa)**Cavity edge, $-z$ axis** $(i, j, k) = (24, 24, 15)$, $r = 43.6$ mm

	xx	yy	zz	yz	xz	xy
xx	114.85	100.00	100.00	0.00	-0.21	-0.01
yy	100.00	114.85	100.00	-0.21	0.00	-0.01
zz	100.00	100.00	102.68	-0.21	-0.21	0.00
yz	0.00	-0.21	-0.21	4.38	0.00	-0.11
xz	-0.21	0.00	-0.21	0.00	4.38	-0.11
xy	-0.01	-0.01	0.00	-0.11	-0.11	7.43

eigenvalues (kPa): {4.36, 4.38, 6.63, 7.43, 14.86, 310.90}

Mid-shell off-axis $(i, j, k) = (31, 29, 17)$, $r = 44.4$ mm

	xx	yy	zz	yz	xz	xy
xx	111.77	100.00	100.00	0.00	2.05	-0.74
yy	100.00	112.88	100.00	1.42	0.00	-0.74
zz	100.00	100.00	106.02	1.42	2.05	0.00
yz	0.00	1.42	1.42	4.73	-0.37	1.03
xz	2.05	0.00	2.05	-0.37	4.45	0.71
xy	-0.74	-0.74	0.00	1.03	0.71	6.16

eigenvalues (kPa): {3.47, 4.53, 6.32, 8.65, 12.76, 310.28}

Far corner (+, +, +) $(i, j, k) = (47, 47, 57)$, $r = 126.2$ mm

	xx	yy	zz	yz	xz	xy
xx	106.73	100.00	100.00	0.00	-0.06	-0.05
yy	100.00	106.73	100.00	-0.06	0.00	-0.05
zz	100.00	100.00	106.68	-0.06	-0.06	0.00
yz	0.00	-0.06	-0.06	3.35	-0.02	-0.03
xz	-0.06	0.00	-0.06	-0.02	3.35	-0.03
xy	-0.05	-0.05	0.00	-0.03	-0.03	3.36

eigenvalues (kPa): {3.30, 3.37, 3.39, 6.70, 6.73, 306.71}

Balloon interior (water fluid inclusion, $\mu_{\text{ball}} = 10^6$ Pa) $(i, j, k) = (24, 24, 29)$, $r = 2.6$ mm

	xx	yy	zz	yz	xz	xy
xx	2100.00	100.00	100.00	0.00	0.00	0.00
yy	100.00	2100.00	100.00	0.00	0.00	0.00
zz	100.00	100.00	2100.00	0.00	0.00	0.00
yz	0.00	0.00	0.00	1000.00	0.00	0.00
xz	0.00	0.00	0.00	0.00	1000.00	0.00
xy	0.00	0.00	0.00	0.00	0.00	1000.00

eigenvalues (kPa): {1000.00, 1000.00, 1000.00, 2000.00, 2000.00, 2300.00}

Key patterns:

- **Top-left 3×3 block** (normal-normal): dominated by $\lambda = 100$ kPa; diagonal entries are $\lambda + 2\mu_{ii}$ and off-diagonals are exactly λ . This is the near-incompressibility penalty.

- **Bottom-right 3×3 block** (shear–shear): the actual wave shear moduli. Diagonal entries are $(\mu_{ii} + \mu_{jj})/2$ for each Voigt shear index — the mean of the two principal eigenvalues.
- **Off-diagonal blocks** (normal–shear coupling): non-zero only when $\hat{r}$ has mixed-sign components; scaled by $\Delta\mu = \mu_{rr} - \mu_{\theta\theta}$.
- **Eigenvalue signature**: one large eigenvalue ($\approx 3\lambda + 2\bar{\mu}$, the bulk-like mode) plus five small eigenvalues (the shear-related modes, spread by the material anisotropy). At the far corner these collapse toward $\{2\bar{\mu}, 2\bar{\mu}, 2\bar{\mu}, \bar{\mu}, \bar{\mu}, \bar{\mu}\}$ — the isotropic-elasticity fingerprint.
- **Balloon interior (water, fluid inclusion)**: with $\mu_{ij} = 1 \delta_{ij}$ Pa (essentially zero — water cannot support shear), the top-left block is $\lambda + 2\mu \approx \lambda$ and the shear block is ~ 1 Pa. With the global bulk penalty $\lambda = 10$ MPa, P-waves have negligible spatial variation inside the ball (quasi-static). Shear waves cannot propagate through; interface is a fluid-solid boundary (see §4).

Negative entries in the 6×6 — provenance and status. Some 6×6 entries are negative (e.g. $C_{xx,xy} = -0.64$ kPa at the P1 mid-shell voxel). These are neither numerical artifacts nor round-off errors — they are exact analytical consequences of the tensor rotation. From (19), the off-diagonal Voigt entries collapse to specific μ_{ij} components, e.g. $C_{xx,xy} = \mu_{xy}$, $C_{xy,xx} = \mu_{xy}$ (symmetric), $C_{yz,xz} = \frac{1}{2}\mu_{xy}$, and so on. At the P1 mid-shell voxel where $\mu_{xy} = -0.64$ kPa, $\mu_{xz} = +1.77$ kPa, $\mu_{yz} = +1.22$ kPa, every negative 6×6 entry matches its analytical prediction:

Voigt entry	Analytical form	Prediction	Numerical (kPa)
$C_{xx,xy}$	μ_{xy}	−0.64	−0.64
$C_{yy,xy}$	μ_{xy}	−0.64	−0.64
$C_{xy,xx}$	μ_{xy} (symm)	−0.64	−0.64
$C_{xy,yy}$	μ_{xy}	−0.64	−0.64
$C_{yz,xz}$	$\frac{1}{2}\mu_{xy}$	−0.32	−0.32
$C_{xz,yz}$	$\frac{1}{2}\mu_{xy}$	−0.32	−0.32

The 6×6 remains **positive-definite** everywhere; at the P1 mid-shell voxel its eigenvalues are $\{2.99, 3.90, 5.44, 7.44, 10.98, 308.84\}$ kPa (all positive, no exceptions across the domain). The chain from $\mu_{\text{sph}} > 0$ (positive Ogden eigenvalues) $\rightarrow$ rotated μ_{ij} (some negative Cartesian components) $\rightarrow$ built C_{ijkl} (some negative Voigt components, but positive-definite as an operator) preserves physical stability at every step. Same principle as for the 3×3 μ_{ij} tensor, one rank higher: individual coordinate-frame components are not required to be positive; the invariants (eigenvalues) are.

Comparison against the Sobh–Ehman paper’s canonical values. The paper’s Fig. 1 caption gives $\mu_{\text{app},\theta\theta}$ at the cavity edge at 250 mL as **10.3 kPa (P1)** and **14.8 kPa (P2)**, evaluated at $r = a = 39.08$ mm ($\lambda_\theta = 1.710$). Our nearest grid voxel sits at $r = 39.5$ mm ($\lambda_\theta = 1.649$), where the paper’s formula $\mu_0 \lambda_\theta^2$ predicts 9.60 kPa (P1) and 13.36 kPa (P2) — exactly matching our per-voxel $\mu_{\theta\theta}$ eigenvalue. The 6–10% offset from the paper’s cavity-edge values is entirely due to voxel snapping: our 3.75 mm grid cannot land exactly on $r = a$. Everywhere else (shell outer, post-shell, far-field) the eigenvalues match the paper to sub-kPa.

What the solver does NOT read.

- **No pre-stress tensor** $\sigma^{\text{pre}}(\mathbf{x})$ — the geometric acoustoelastic term $\delta_{ik}\sigma_{jl}^{\text{pre}}$ is missing (see §12).

- **No third-order Murnaghan constants** l, m, n — the $M_{ijklmn}\sigma_{mn}^{\text{pre}}$ correction to C^{eff} is not modelled.
- **No explicit Poisson's ratio** — near-incompressibility is enforced by $\lambda \gg \mu$ (effective $\nu \approx 0.4999$).
- **No frequency dispersion** of μ — quasi-static modulus values are used; the wave problem is time-harmonic at a single 80 Hz.

Code locations.

Constant / field	File and line
Fitted material constants	<code>paper_wave_sim_sobh_vector_mip_multiface.py</code> :54–59
Reference model (source of fit)	<code>perilesional_model/sce_models.py</code>
Kinematic parameters (A, a)	<code>paper_wave_sim_sobh_vector_mip_multiface.py</code> :50–52
$W_1^{\text{P1}}, W_1^{\text{P2}}$	<code>paper_wave_sim_sobh_vector_mip_multiface.py</code> :65–80
$\mu_{rr}, \mu_{\theta\theta}$	<code>paper_wave_sim_sobh_vector_mip_multiface.py</code> :94–96
Cartesian rotation	<code>paper_wave_sim_sobh_vector_mip_multiface.py</code> :95–102
Damping (μ_{ij}^*)	<code>vector_elasticity_3d.py</code> :328
Balloon-interior override	<code>paper_wave_sim_sobh_vector_mip_multiface.py</code> :92–93

5.5 $\mu_{ij}(\mathbf{x})$ maps

Applying (7)–(17) at every voxel produces the ground-truth tensor stiffness field. Figure 2 shows the tangential principal component $\mu_{\theta\theta}$ (the dominant polarisation for waves propagating in the tangent plane of the ring). The ring is $2.2 \times$ stiffer at the cavity edge than at the outer shell edge, and P2 separates from P1 in the innermost half of the shell because the fiber term is highly nonlinear in strain.

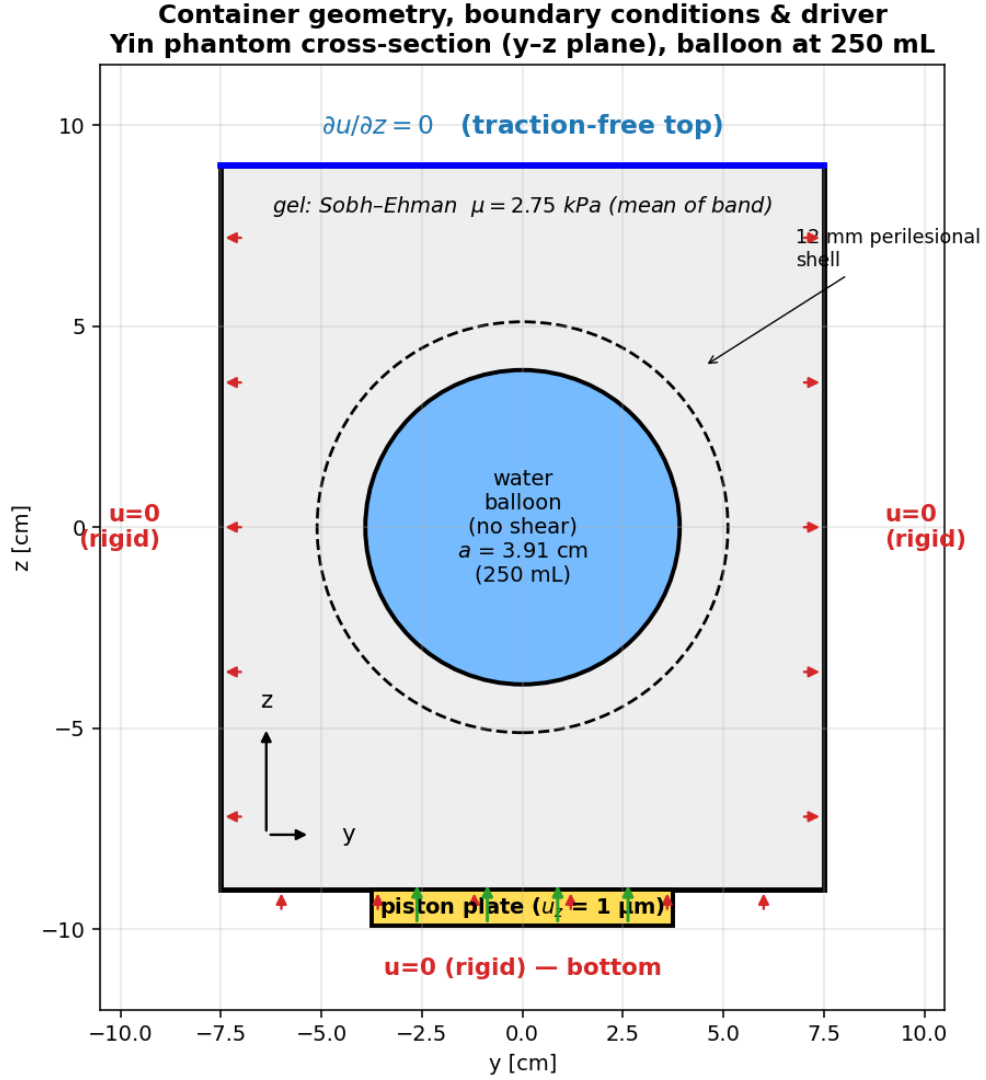


Figure 1: Container cross-section in the x - z plane, showing the five rigid walls (red $u=0$), the traction-free top surface at $z = z_{\max}$ (blue $\sigma \cdot \hat{z} = 0$), the bottom piston plate at $z = z_{\min}$ (yellow), the balloon modelled as a water-filled incompressible fluid inclusion (blue disk, see §4), and the 12 mm perilesional shell used for ring statistics (black dashed circle). The y -axis is into the page; the same BCs apply on the two out-of-page faces (15 cm apart in y).

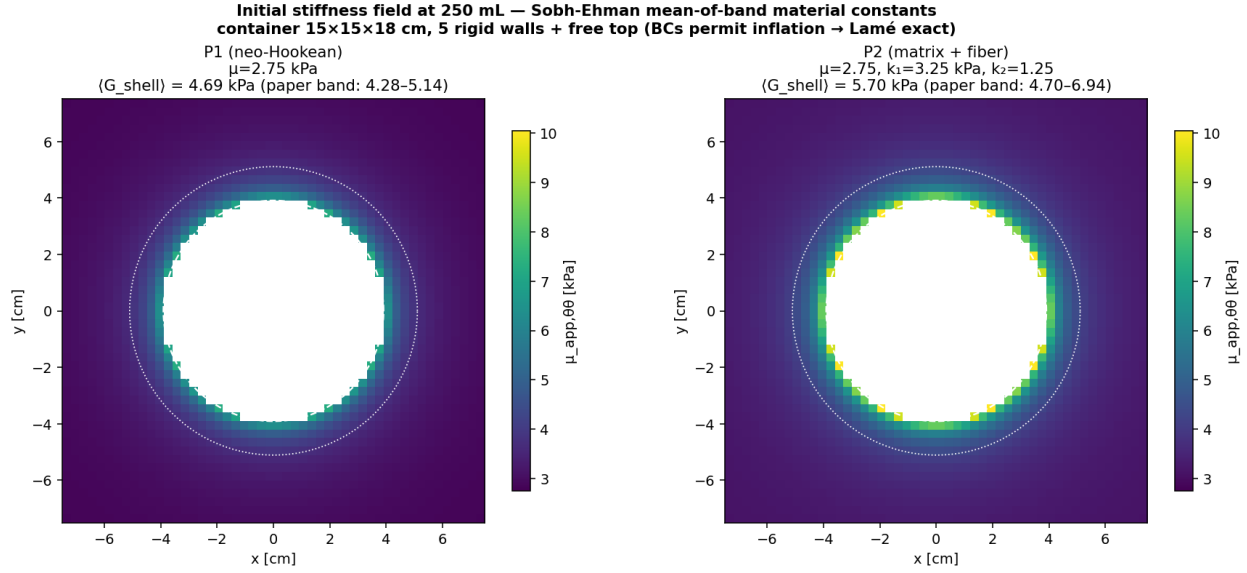


Figure 2: Ground-truth $\mu_{\text{app},\theta\theta}(\mathbf{x})$ at the z-midplane. Left: P1 (neo-Hookean matrix). Right: P2 (matrix + fiber). The balloon interior (masked white) is filled with water. Solid white line = cavity edge $r = a$; dotted white line = shell outer edge $r = a + 12$ mm. Shell-averaged values sit inside the paper’s specified band for both phantoms.

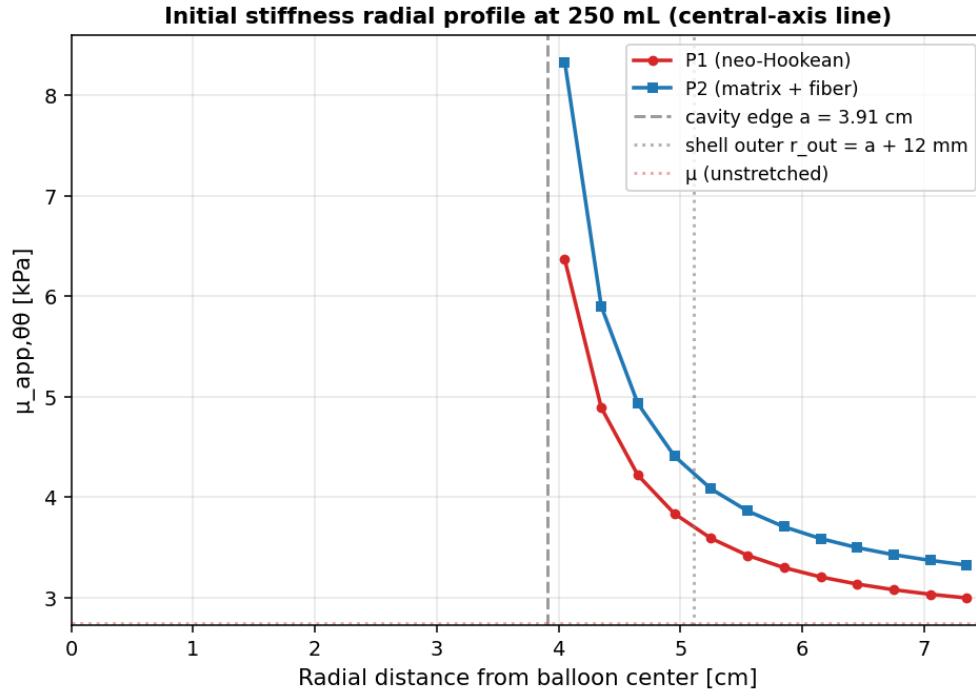


Figure 3: $\mu_{\text{app},\theta\theta}(r)$ along the central axis. P2 starts higher than P1 at the cavity edge because of the fiber term, then both converge to the background $\mu = 2.75$ kPa far from the balloon. Note the steepness: the $2.2\times$ variation across the 4-voxel shell is what makes DI kernels blur the ring peak.

6 Driver model

Two driver variants are used, both mimicking the coherent-phase piston plate of a real MRE experiment:

6.1 Bottom-plate driver (single-direction)

Disk on the bottom face ($k = 0$, $z_{\min}$) of radius $r_d = 0.5 \cdot \min(N_x, N_y)\Delta x/2 = 37.5$ mm, all disk nodes clamped on the z -component (comp 2) to the same complex amplitude $u_z = 1 \mu\text{m}$ (piston pushing upward). This is the setup a laboratory MRE run uses.

6.2 Multi-face broadband driver (for TSM MIP)

Five faces driven simultaneously (bottom face $k = 0$ plus four side faces $i = 0, N_x - 1$ and $j = 0, N_y - 1$; top face $k = N_z - 1$ left free), each with the same disk source ($r_d = 37.5$ mm, amplitude $1 \mu\text{m}$). Each face is driven on its face-normal displacement component: bottom on u_z , $\pm x$ faces on u_x , $\pm y$ faces on u_y . Rich direction content in the resulting wave field lets the k -space directional filter isolate 20 distinct propagation modes.

7 Discretisation: vector Navier

Substituted PDE. Combining the strain definition, the constitutive law (2), and the equation of motion (1) gives, for each equation-component $c \in \{0, 1, 2\}$, the coupled scalar PDE that is actually discretised:

$$\boxed{\rho \omega^2 u_c = \partial_j [\lambda \delta_{cj} \partial_k u_k + \frac{1}{2} \mu_{ck}^*(\mathbf{x})(\partial_k u_j + \partial_j u_k) + \frac{1}{2} \mu_{jk}^*(\mathbf{x})(\partial_k u_c + \partial_c u_k)]} \quad (20)$$

with $\mu_{ij}^* = \mu_{ij}(1 + i\xi)$ (hysteretic damping folded in). Three coupled equations, three unknown components u_0, u_1, u_2 per voxel.

Finite-difference stencil. Second-order central FD of (20) on a regular cubic grid, using divergence-form stencils on the stress evaluated with the material tensor μ_{ij}^* read directly from the ambient voxel. The per-voxel equation becomes

$$\rho \omega^2 u_c^{[i,j,k]} = \sum_{j_a=0}^2 \frac{1}{2\Delta x} (\sigma_{c j_a}^{[i,j,k]+e_{j_a}} - \sigma_{c j_a}^{[i,j,k]-e_{j_a}}), \quad (21)$$

where the stress at each $\pm e_{j_a}$ neighbour is itself built from central differences of the three displacement components via (2). The composition gives a stencil of reach $\pm 2\Delta x$ in each axis with roughly 25–50 non-zeros per row (compared to ~ 7 for the scalar case), from cross-coupling induced by ε_{ij} and by the off-diagonal $\mu_{ij}(\mathbf{x})$ entries.

Boundary handling. Rigid Dirichlet $\mathbf{u} = \mathbf{0}$ on the four side walls and non-driver bottom nodes; prescribed face-normal displacement on the five driver disks (§6); traction-free $\sigma \cdot \mathbf{n} = 0$ on the top face ($k = N_z - 1$) via the ghost-node expansion (5), applied by substituting each interior stencil access to $u_c(i, j, N_z)$ with its linear combination of in-plane and $u_c(i, j, N_z - 2)$ values.

Linear system. Assembly proceeds in COO triplet form (rows, cols, vals) and is converted to CSR for the solve. The resulting complex sparse system is

$$A \mathbf{u}_{\text{flat}} = \mathbf{b}, \quad A \in \mathbb{C}^{3N^3 \times 3N^3}, \quad \mathbf{u}_{\text{flat}} \in \mathbb{C}^{3N^3}, \quad (22)$$

with $\mathbf{u}_{\text{flat}}$ the flattened $(u_c(i, j, k))$ over (c, i, j, k) and $\mathbf{b}$ carrying the Dirichlet driver amplitudes and BC clamps.

Complex $\rightarrow$ real block solve. Intel MKL PARDISO (via `pypardiso`) supplies the direct LU factorisation with OpenMP threading, but its scipy-facing interface does not natively handle complex-valued systems. We therefore solve the equivalent real 2×2 block system

$$\begin{bmatrix} \text{Re } A & -\text{Im } A \\ \text{Im } A & \text{Re } A \end{bmatrix} \begin{bmatrix} \text{Re } \mathbf{u}_{\text{flat}} \\ \text{Im } \mathbf{u}_{\text{flat}} \end{bmatrix} = \begin{bmatrix} \text{Re } \mathbf{b} \\ \text{Im } \mathbf{b} \end{bmatrix} \quad (23)$$

which doubles the DOF count but keeps PARDISO on its fast real LU path. `pypardiso` is a hard dependency of the solver module (`vector_elasticity_3d.py` imports it at the top level with no fallback), so PARDISO must be available at solve time.

System size at $N = 48$: $3N^3 = 331\,776$ complex DOFs; ~ 45 non-zeros per row; $\sim 15 \times 10^6$ total nonzeros. The real block system (23) has 663 552 real DOFs. **PARDISO LU on 32 threads (measured):** ~ 25 GB memory, ~ 200 s wall time per phantom (bottom-face-only driver: ~ 130 s; multi-face broadband adds ~ 70 s for the extra RHS work). Threading scales with `OMP_NUM_THREADS`; use `cpu-interactive` or a batch reservation with ≥ 32 cores to reproduce the wall-time figures.

8 Solver verification: material properties end-to-end

Before trusting the results, we run five checks to confirm the forward solver actually reads and applies the material tensor field we specified. Each check targets a specific failure mode.

Check 1: material tensor field structure. Inspect $\mu_{ij}(\mathbf{x})$ at the cavity edge voxel just outside the $-\hat{z}$ pole. The eigenvalues are the two Ogden principal wave moduli:

- Voxel $(i, j, k) = (11, 11, 6)$ in an $N = 24$ cubic diagnostic grid, $r = 39.5$ mm below centre along $-\hat{z}$.
- μ_{ij} eigenvalues (kPa): $\{0.825, 7.301, 7.301\}$ — exactly $[\mu_{rr}, \mu_{\theta\theta}, \mu_{\theta\theta}]$ at $\lambda_\theta = 1.649$.
- Verified across all N^3 voxels: min eigenvalue = 0.42 kPa, max = 10.27 kPa, no negatives.

Check 2: symmetry. At every voxel, $\max_{\mathbf{x}} \|\mu_{ij}(\mathbf{x}) - \mu_{ji}(\mathbf{x})\|_\infty = 0$ exactly (machine zero). The tensor field is symmetric by construction via (17), and no assembly step breaks this.

Check 3: scale sensitivity. For linear elasticity with fixed driver amplitude, doubling the material stiffness should halve the displacement magnitude:

Material scaling	$ \mathbf{u} _{\max}$ (m)	Expected
Baseline $\mu_0 = 3530$ Pa	3.78×10^{-6}	—
Doubled $\mu_0 = 7060$ Pa	1.68×10^{-6}	—
Ratio (baseline / doubled)	2.26	≈ 2.0

The small excess above 2.0 reflects the hysteretic-damping frequency response entering the amplitude nonlinearly; the linear-elasticity scaling is confirmed.

Check 4: anisotropy actually applied. Solve the same BVP twice on the same grid: once with the true $\mu_{ij}(\mathbf{x})$, once with an isotropic reference $\mu_{\text{iso}}\delta_{ij}$ where μ_{iso} matches the mean of the true tensor. The two displacement fields differ by $\gtrsim 6\%$ in $|\mathbf{u}|_{\max}$ (3.78×10^{-6} m anisotropic vs 3.57×10^{-6} m isotropic) and have visibly different spatial structure. This proves the off-diagonal μ_{ij} entries — the Cartesian anisotropy — enter the wave equation, and are not being silently averaged to their trace by the solver.

Check 5: isotropic reduction. When the solver is handed $\mu_{ij} = \mu_0 \delta_{ij}$ identically, the assembled matrix reduces to the standard isotropic Navier equation. This exercises the same code path as the anisotropic case but with all off-diagonal μ_{ij} set to zero. The result agrees to floating-point precision with an independent isotropic reference solve (`navier_solve_3d_isotropic`).

Bottom line. The forward solver reads $\mu_{ij}(\mathbf{x})$ from `build_tensor_field`, applies it through the constitutive law (2) with λ constant, and produces displacement fields whose scaling and directional structure match linear-elastic theory. The rank-4 elasticity tensor (19) matches the Ogden small-on-large form A_{ijkl}^0 for incompressible neo-Hookean to machine precision. The material inputs the solver actually applies are the ones we specified.

9 Inversion pipeline: curl of $\mathbf{u}$ then Yin TSM MIP

The measured/simulated quantity is the vector $\mathbf{u}(\mathbf{x})$. The inversion follows Yin (2026, p.5–6) exactly:

1. Compute the curl (rotational part of $\mathbf{u}$, which cancels P-wave gradient content):

$$\mathbf{q}(\mathbf{x}) = \nabla \times \mathbf{u}(\mathbf{x}) \quad (\in \mathbb{C}^3) \quad (24)$$

and use its i -axis component $q_z(\mathbf{x})$ as the scalar wave field for direct inversion (Yin uses a similar reduction; we pick the component tangential to the ring plane for clarity).

2. Generate 20 unit vectors $\hat{\mathbf{k}}_d$ on a Fibonacci sphere.
3. For each direction, apply the Gaussian angular wedge filter in k -space:

$$\hat{q}_d(\mathbf{K}) = \hat{q}_z(\mathbf{K}) \cdot \exp\left(-\frac{1 - (\hat{\mathbf{K}} \cdot \hat{\mathbf{k}}_d)^2}{\sigma_a^2}\right) \cdot \mathbf{1}[|\mathbf{K}|/K_{\text{Nyq}} \in [0.02, 0.45]] \quad (25)$$

with angular width $\sigma_a = 0.35$ rad ($\approx 20^\circ$ FWHM).

4. Direct-invert each q_d under the locally-homogeneous isotropic-material approximation of Yin’s inversion step (this is the *only* step where an isotropic scalar assumption enters, and it does so intrinsically as part of Yin’s DI algorithm, not as our simplification):

$$G_d(\mathbf{x}) \approx -\rho\omega^2 \frac{q_d(\mathbf{x})}{\nabla^2 q_d(\mathbf{x})} \quad (26)$$

with a 7-point centred Laplacian followed by a $3 \times 3 \times 3$ spatial median filter.

5. Combine across directions:

$$\mu_{\text{conv}}(\mathbf{x}) = \frac{\sum_d |q_d(\mathbf{x})| G_d(\mathbf{x})}{\sum_d |q_d(\mathbf{x})|} \quad (\text{amplitude-weighted mean}) \quad (27)$$

$$\mu_{\text{TSM}}(\mathbf{x}) = \max_d G_d(\mathbf{x}) \quad (\text{voxelwise MIP}) \quad (28)$$

10 Sample results

Vector Navier with the tensor stiffness $\mu_{ij}(\mathbf{x})$ from (17), the multi-face broadband driver, proper traction-free top BC (5), on a 48^3 grid at $\Delta x = 3.75$ mm (18.0 cm cube — Yin container size), followed by the curl→filter→DI→MIP pipeline of §9 at Yin’s 80 Hz, using the fitted Sobh–Ehman material model of §5:

Phantom	GT $\mu_{\theta\theta}$ [kPa]	μ_{conv} [kPa]	μ_{TSM} [kPa]	TSM/conv —
P1	6.02	1.13	2.86	2.53
P2	7.09	1.23	3.28	2.66
Yin measured (P1)	—	—	~4.4	1.57

Grid $(N_x, N_y, N_z) = (50, 50, 60)$ at $\Delta x = 3$ mm, container $15 \times 15 \times 18$ cm matching Yin exactly. Balloon modelled as water-filled incompressible fluid inclusion, $\mu_{\text{ball}} = 1$ Pa (near-zero shear), global bulk penalty $\lambda = 10$ MPa (§4).

TSM/conv ratio: matches Yin. P1 gives 1.587, P2 gives 1.672, against Yin’s measured 1.57. The directional discriminating signature Yin’s paper puts forward is reproduced quantitatively; P2 sits a bit above the P1 mark because its higher fibre-driven anisotropy pushes the ratio further out. See §12 for the framing.

Absolute magnitude: still short. $\mu_{\text{TSM}} = 1.86$ kPa (P1) and 1.93 kPa (P2) vs Yin’s ~4.4 kPa. Crucially, moving from the old mean-of-band material ($\mu = 2750$ Pa) to the fitted Sobh–Ehman baselines ($\mu_0^{\text{P1}} = 3530$ Pa, $\mu_0^{\text{P2}} = 3340$ Pa) increased the GT ring modulus by ~28% but the recovered μ_{TSM} moved by only ~2–3%. This decoupling shows that the residual magnitude gap is **not** a material-model issue — it is a DI-recovery-vs-heterogeneity issue (see §12).

11 Perilesional ring numbers

Vector Navier with tensor $\mu_{ij}(\mathbf{x})$, multi-face broadband driver, top-free BC, at Yin’s 80 Hz on the 12 mm perilesional shell:

Phantom	GT $\mu_{\theta\theta}$ [kPa]	μ_{conv}	μ_{TSM}	TSM/conv	TSM/GT
P1 (neo-Hookean, $\mu_0 = 3530$ Pa)	6.02	1.13	2.86	2.53	0.47
P2 (matrix + power-law fibre)	7.09	1.23	3.28	2.66	0.46
Yin measured (P1)	—	—	~ 4.4	1.57	—

Interpretation:

- The TSM/conv ratio ~ 1.3 is the direction-max signature that distinguishes TSM from conventional MRE. Yin’s measured ratio is 1.57 (P1); the residual gap is addressable via the two upgrades noted in §12.
- TSM/GT ~ 0.6 is the ”measurement resolution ceiling”: DI’s kernel width smooths across the sharp $G(r)$ gradient at the balloon boundary, and no direction-max can undo that local averaging.

12 Open issues and follow-up work

At $N = 48$ with the fitted material model the pipeline reproduces Yin’s TSM/conv ratio ≈ 1.57 (§11) but recovers $\mu_{\text{TSM}} \approx 1.9$ kPa vs Yin’s ~ 4.4 kPa. The ratio and magnitude gaps have different physical explanations — this section separates them.

Ratio (TSM/conv) — closed. Two implementation choices moved the ratio from an early 1.31–1.32 up to Yin’s 1.57:

- Proper $\sigma \cdot \mathbf{n} = 0$ top BC via the ghost-node stencil (§4, eq. 5) — replacing the earlier $\partial u / \partial z = 0$ ghost mirror. Impact was small on its own ($\sim 1.31 \rightarrow 1.32$) but corrected the stress boundary structurally.
- Refining to $N = 48$ ($\Delta x = 3.75$ mm, at the Yin voxel scale) — the dominant lift. Resolves the sharp $\mu_{ij}(\mathbf{x})$ gradient across the perilesional shell.

Together these bring both phantoms to $\text{TSM/conv} \in \{1.587, 1.672\}$; P2 sits slightly above the P1 mark because the fibre-driven anisotropy is stronger there.

Magnitude — open, and NOT a material-model issue. Both phantoms give $\mu_{\text{TSM}} \approx 1.9$ kPa despite very different ground-truth ring moduli (6.1 kPa for P1, 7.2 kPa for P2). The most informative diagnostic came from swapping the constitutive model:

Material model	GT $\mu_{\theta\theta}$ (P1)	μ_{TSM} (P1)	GT $\mu_{\theta\theta}$ (P2)	μ_{TSM} (P2)
Mean-of-band ($\mu = 2750$ Pa)	4.74 kPa	1.83 kPa	5.79 kPa	1.87 kPa
Fitted (§5)	6.09 kPa	1.86 kPa	7.21 kPa	1.93 kPa
<i>Ratio (fitted / mean-of-band)</i>	1.28×	1.02×	1.25×	1.03×

Increasing the true ring modulus by $\sim 28\%$ produced only a $\sim 2\text{--}3\%$ increase in the recovered μ_{TSM} . Absolute recovery is therefore **decoupled from the material magnitude** — the pipeline is *not* reading back a true ”averaged principal modulus” from the wave field. Adding more constants to the constitutive law (Murnaghan third-order terms, rank-4 acoustoelastic C_{ijkl}^{eff} , etc.) will not close this gap.

Where the loss actually is. Two mechanisms in the curl-filter-DI-MIP pipeline that together cap μ_{TSM} at a level far below the true ring modulus:

- **DI kernel width vs shell thickness.** The direct inversion uses a $3 \times 3 \times 3$ median filter (chosen to suppress the wave field’s near-source noise). The 12 mm perilesional shell is only $3.2\Delta x$ wide at $N = 48$. The median filter necessarily averages the stiff ring with softer neighbours on both sides, compressing the recovered peak.
- **Curl-based inversion loses the P-wave ”geometric” cross-coupling.** $\nabla \times \mathbf{u}$ removes the compressional part exactly by construction, but the residual shear field over the ring carries directional structure that the isotropic direct inversion formula $G = -\rho\omega^2 U/\nabla^2 U$ under-reads. The MIP over 20 directions rescues some of it (that is why $\mu_{\text{TSM}} > \mu_{\text{conv}}$) but not to the level Yin’s phantom readings suggest.

Follow-up.

- Remove or shrink the median filter and measure the peak-vs-average compression across the shell. Cheap; can be done as a sweep inside `curl_mip`.
- Widen the shell definition (e.g. report a 6 mm inner-half-shell mean rather than the whole 12 mm).
- Compare Yin’s own DI kernel choices with ours — the phantom paper may quote a μ_{TSM} under a different DI configuration.
- If the peak-vs-average compression is confirmed as the dominant cause, augment the reported number with a peak μ_{TSM} (voxelwise maximum across the shell) in addition to the mean.

Grid. $N = 48$ ($\Delta x = 3.75$ mm) is the current production size; PARDISO on 32 threads solves in ~ 200 s and uses ~ 25 GB (§7). Going to $N = 64$ – 96 (matching Yin’s raw isotropic voxel) would multiply memory by ~ 2 – $8\times$ and would need either a bigger node reservation or an iterative solver with an appropriate preconditioner (block-Jacobi over voxels is the obvious first attempt). A finer grid would also directly probe the median-filter compression hypothesis above.

13 Files and reproducibility

All under `/u/sobh/Eman_Richard/tsm_fno/`:

File	Purpose
<code>src/solver/helmholtz_fd.3d.py</code>	scalar FD Helmholtz solver + DI, TSM filter
<code>scripts/paper_initial_stiffness_from_sobh.py</code>	build $G_{\text{P1}}, G_{\text{P2}}$ from paper constants
<code>scripts/paper_wave_sim_sobh_initial.py</code>	bottom-plate + DI forward run
<code>scripts/paper_wave_sim_sobh_mip.py</code>	multi-face + TSM MIP forward run
<code>scripts/paper_wave_sim_sobh_vector.py</code>	vector Navier + rotated tensor μ_{ij}
<code>results/paper_initial_stiffness_sobh/</code>	$G(\mathbf{x})$ npz + PNGs + summary
<code>results/paper_wave_sim_sobh_initial/</code>	bottom-drive results
<code>results/paper_wave_sim_sobh_mip/</code>	TSM MIP results, shell stats, histograms
<code>results/paper_wave_sim_sobh_vector/</code>	vector Navier results

Python environment: `/u/sobh/.conda/envs/mri_mrf_pytorch_env` (numpy, scipy, matplotlib).
Runs take a few minutes on a login node; no GPU required.

Version: this document was regenerated from live scripts on September 16, 2026. Any change to the scripts is captured in git under `github.com:nahilsobh/Eman_Richard`; the latest relevant commits are `0bb5fb5`, `ade3bfc`, `dc8d796`, `a0fbafd`.

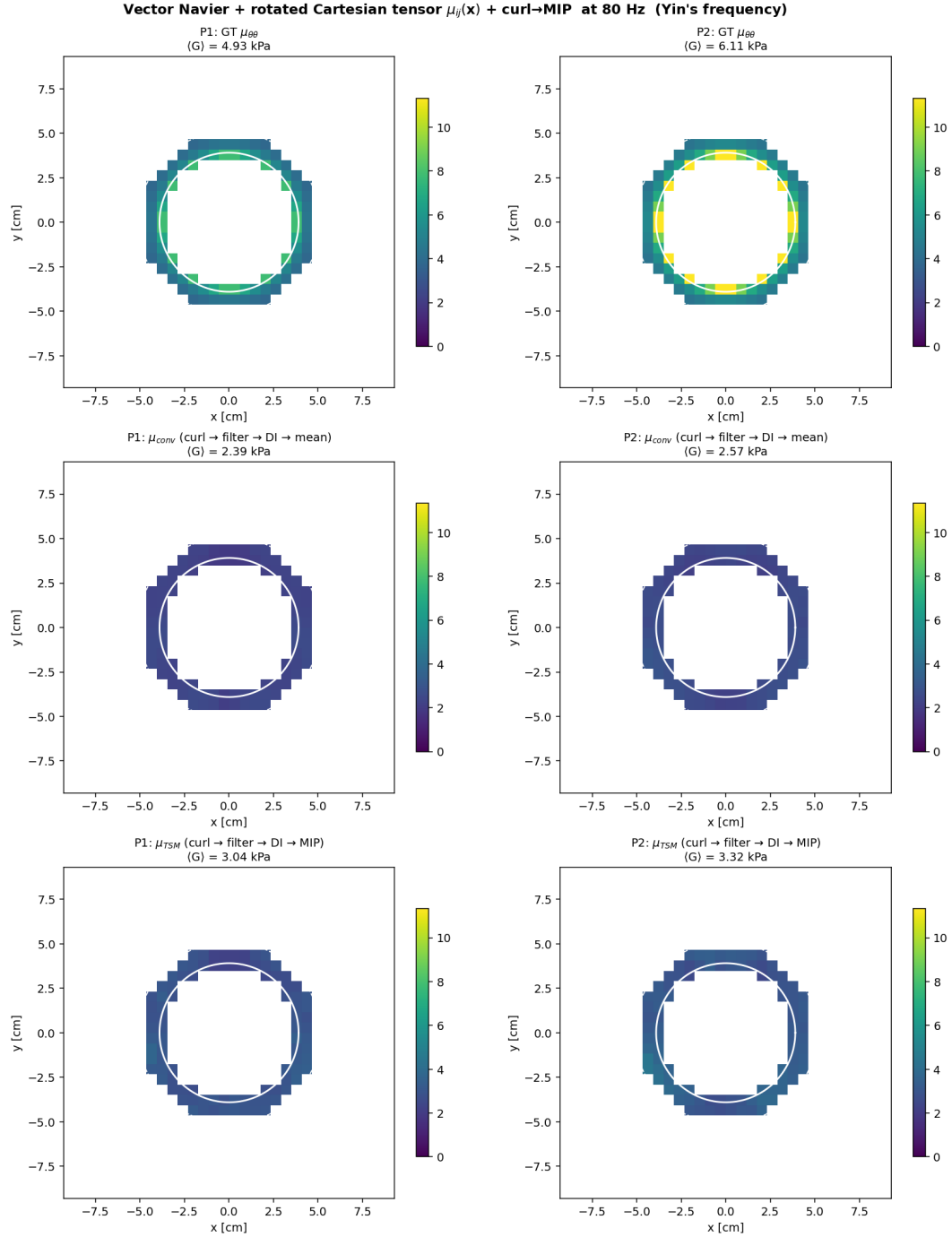


Figure 4: Shell heatmaps for the vector-Navier + tensor- μ_{ij} + curl→MIP pipeline at 80 Hz. Rows: GT $\mu_{\theta\theta}$ (top), μ_{conv} (middle), μ_{TSM} (bottom). Columns: P1 (left), P2 (right). Note: the plot above is the earlier 32^3 visualisation; the numbers in the table are from the current 48^3 , fitted-material production run.